

MERN STACK

DAY – 11

7 July 2026

JAVASCRIPT

Document Object Model

The Document Object Model (DOM) is a programming interface that represents an HTML document as a tree-like structure. Every HTML element, attribute, and piece of text is treated as an object (node). JavaScript uses the DOM to access, modify, add, delete, and style HTML elements dynamically without reloading the web page.

When a browser loads an HTML page, it creates the DOM automatically. JavaScript communicates with the web page through this DOM.

Accessing Elements in DOM

JavaScript uses the document object to access HTML elements.

Syntax:

```
document.methodName("selector");
```

1. getElementById()

Selects a single element using its id.

Syntax:

```
document.getElementById("idName");
```

2. getElementsByClassName()

Returns all elements having the given class name.

Syntax:

```
document.getElementsByClassName("className");
```

3. getElementsByTagName()

Selects all elements with the specified HTML tag.

Syntax:

```
document.getElementsByTagName("tagName");
```

4. querySelector()

Returns the first matching element using a CSS selector.

Syntax:

```
document.querySelector("CSS Selector");
```

5. querySelectorAll()

Returns all matching elements using CSS selectors.

Syntax:

```
document.querySelectorAll("CSS Selector");
```